



Chapter 14: Indexing

Database System Concepts, 7th Ed.

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Outline

- Basic Concepts
- Ordered Indices
- B⁺-Tree Index Files
- B-Tree Index Files
- Hashing
- Write-optimized indices
- Spatio-Temporal Indexing



Basic Concepts

- Indexing mechanisms used to speed up access to desired data.
 - E.g., author catalog in library
- **Search Key** - set of attributes used to look up records in a file.
- An **index file** consists of **index entries** of the form

search-key	pointer
------------	---------

- Index files are typically much smaller than the original file
- Two basic kinds of indices:
 - **Ordered indices:** search keys are stored in sorted order, e.g., B+ tree
 - **Hash indices:** search keys are distributed uniformly across “buckets” using a “hash function”.



Index Evaluation Metrics

- Access types: e.g.,
 - Records with a specified value in the attribute
 - Records with an attribute value falling in a specified range of values.
- Access time
- Insertion time
- Deletion time
- Space overhead



Ordered Indices

- In an **ordered index**, index entries are stored sorted on the search key value.
- **Clustering index**: in a sequentially ordered file, the index whose search key specifies the sequential order of the file.
 - Also called **primary index**
 - The search key of a primary index is usually but not necessarily the primary key.
- **Secondary index**: an index whose search key specifies an order different from the sequential order of the file. Also called **nonclustering index**.
- **Index-sequential file**: sequential file ordered on a search key, with a clustering index on the search key.



Dense Index Files

- **Dense index** — Index record appears for every search-key value in the file.
- E.g. index on *ID* attribute of *instructor* relation

10101		→	10101	Srinivasan	Comp. Sci.	65000	
12121		→	12121	Wu	Finance	90000	
15151		→	15151	Mozart	Music	40000	
22222		→	22222	Einstein	Physics	95000	
32343		→	32343	El Said	History	60000	
33456		→	33456	Gold	Physics	87000	
45565		→	45565	Katz	Comp. Sci.	75000	
58583		→	58583	Califieri	History	62000	
76543		→	76543	Singh	Finance	80000	
76766		→	76766	Crick	Biology	72000	
83821		→	83821	Brandt	Comp. Sci.	92000	
98345		→	98345	Kim	Elec. Eng.	80000	



Dense Index Files (Cont.)

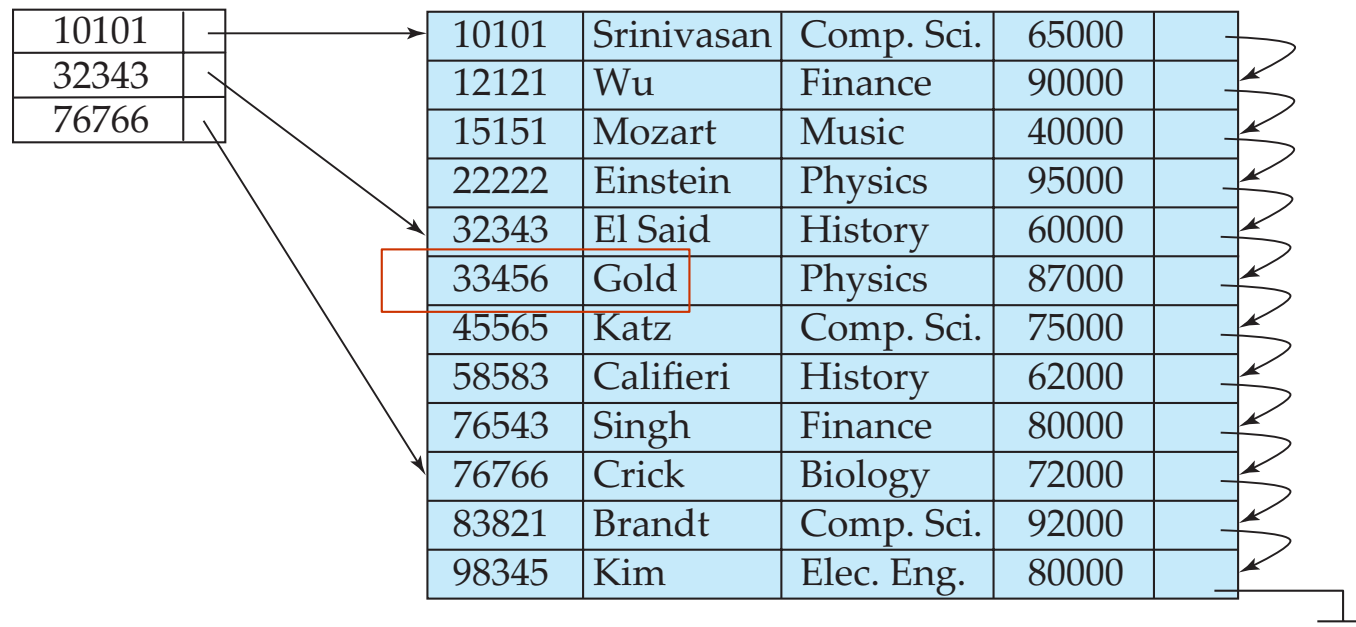
- Dense index on *dept_name*, with *instructor* file sorted on *dept_name*

Biology		76766	Crick	Biology	72000	
Comp. Sci.		10101	Srinivasan	Comp. Sci.	65000	
Elec. Eng.		45565	Katz	Comp. Sci.	75000	
Finance		83821	Brandt	Comp. Sci.	92000	
History		98345	Kim	Elec. Eng.	80000	
Music		12121	Wu	Finance	90000	
Physics		76543	Singh	Finance	80000	
		32343	El Said	History	60000	
		58583	Califieri	History	62000	
		15151	Mozart	Music	40000	
		22222	Einstein	Physics	95000	
		33465	Gold	Physics	87000	



Sparse Index Files

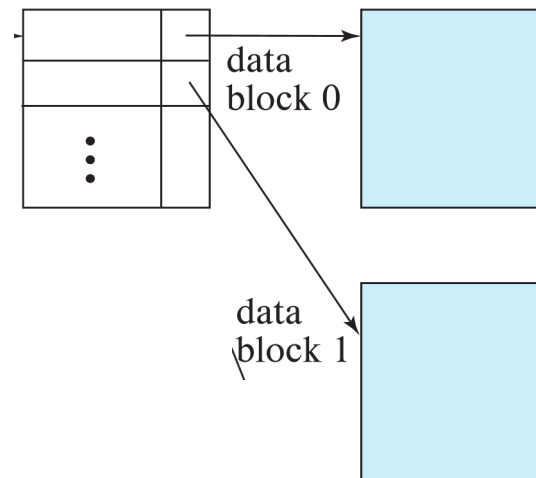
- **Sparse Index:** contains index records for only some search-key values.
 - Applicable when records are sequentially ordered on search-key
- To locate a record with search-key value K we:
 - Find index record with largest search-key value $< K$
 - Search file sequentially starting at the record to which the index record points





Sparse Index Files (Cont.)

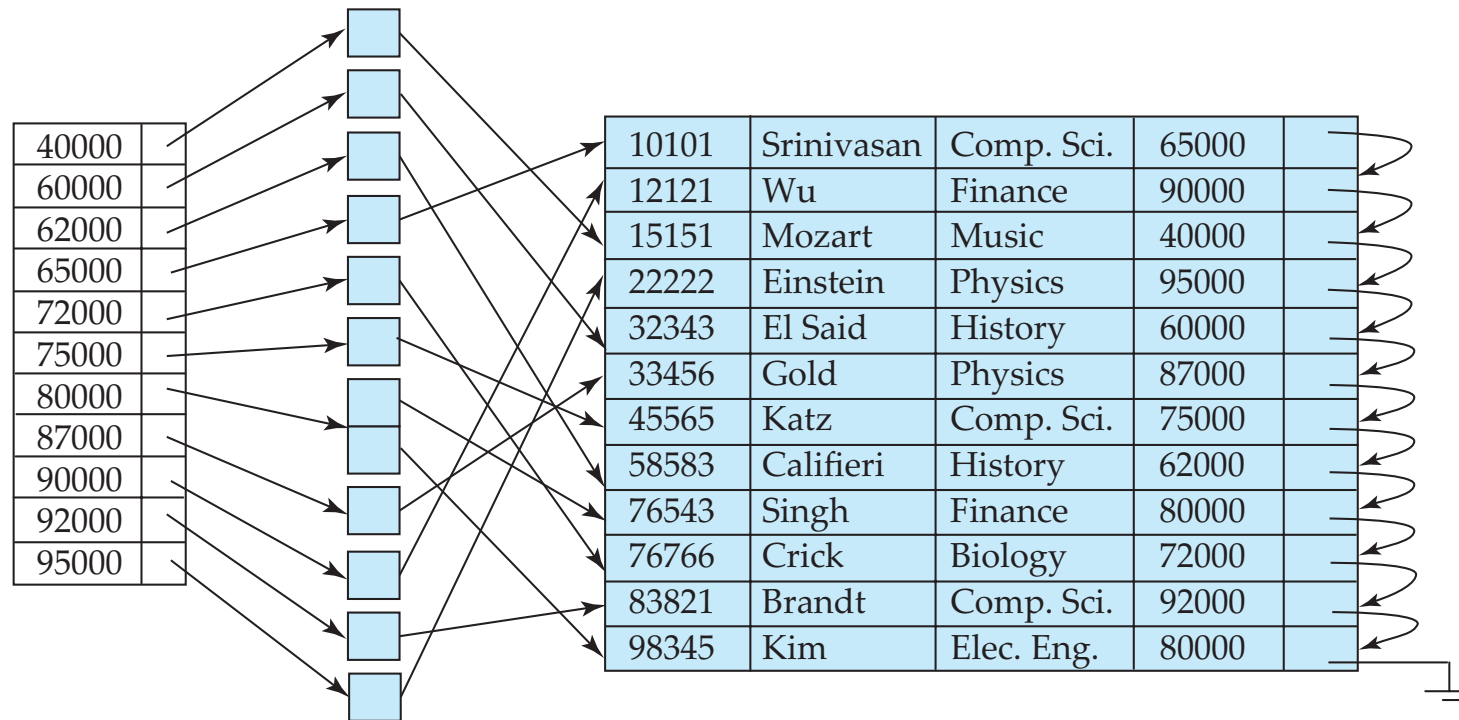
- Compared to dense indices:
 - Less space and less maintenance overhead for insertions and deletions.
 - Generally slower than dense index for locating records.
- **Good tradeoff:**
 - **For clustered index:** sparse index with an index entry for every block in file, corresponding to least search-key value in the block.





Secondary (un-clustered) Indices Example

- Secondary index on **salary** field of instructor



- Index record points to a bucket that contains pointers to all the actual records with that particular search-key value.
- Secondary indices have to be dense



Clustering vs Nonclustering Indices

- Indices offer substantial benefits when searching for records.
- BUT: indices imposes overhead on database modification
 - when a record is inserted or deleted, every index on the relation must be updated
 - When a record is updated, any index on an updated attribute must be updated
- Sequential scan using clustering index is efficient, but a sequential scan using a secondary (nonclustering) index is expensive on magnetic disk
 - Each record access may fetch a new block from disk
 - Each block fetch on magnetic disk requires about 5 to 10 milliseconds

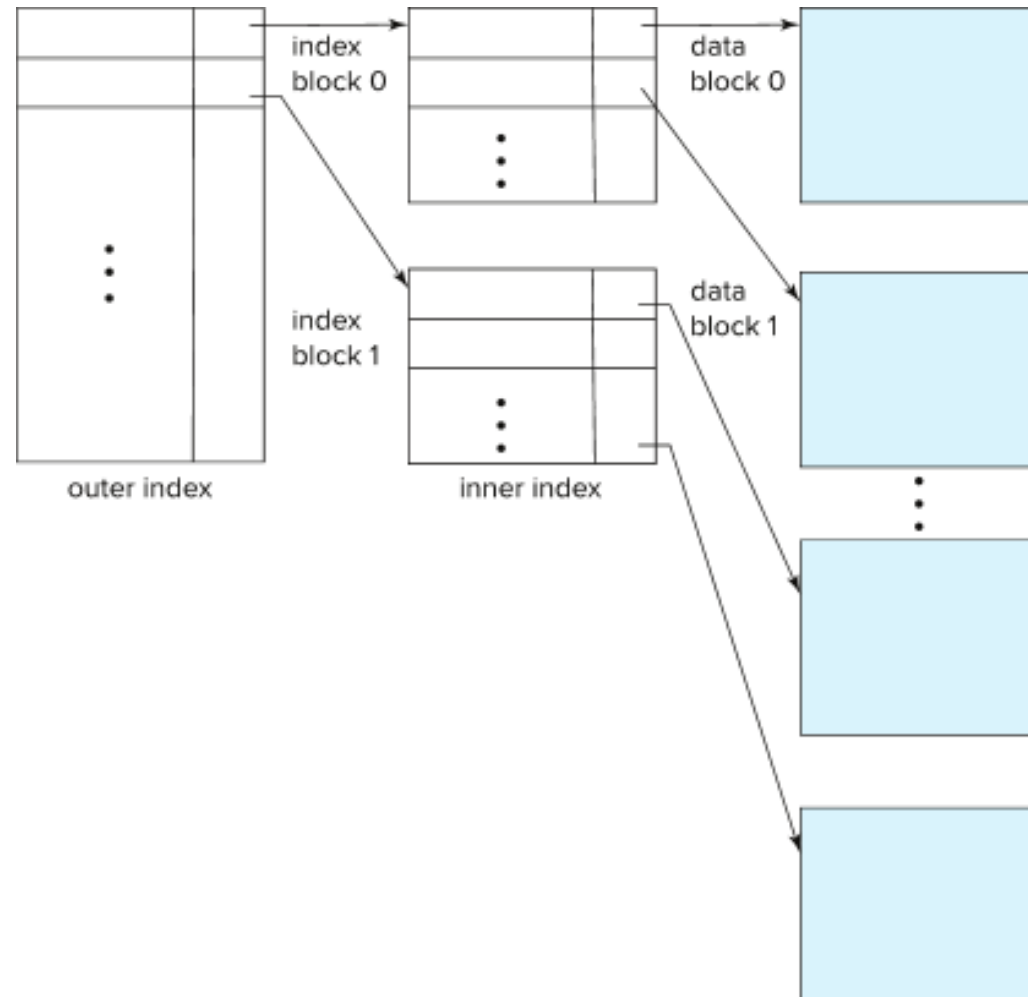


Multilevel Index

- If index does not fit in memory, access becomes expensive.
- Solution: treat index kept on disk as a sequential file and construct a sparse index on it.
 - outer index – a sparse index of the basic index
 - inner index – the basic index file
- If even outer index is too large to fit in main memory, yet another level of index can be created, and so on.
- Indices at all levels must be updated on insertion or deletion from the file.

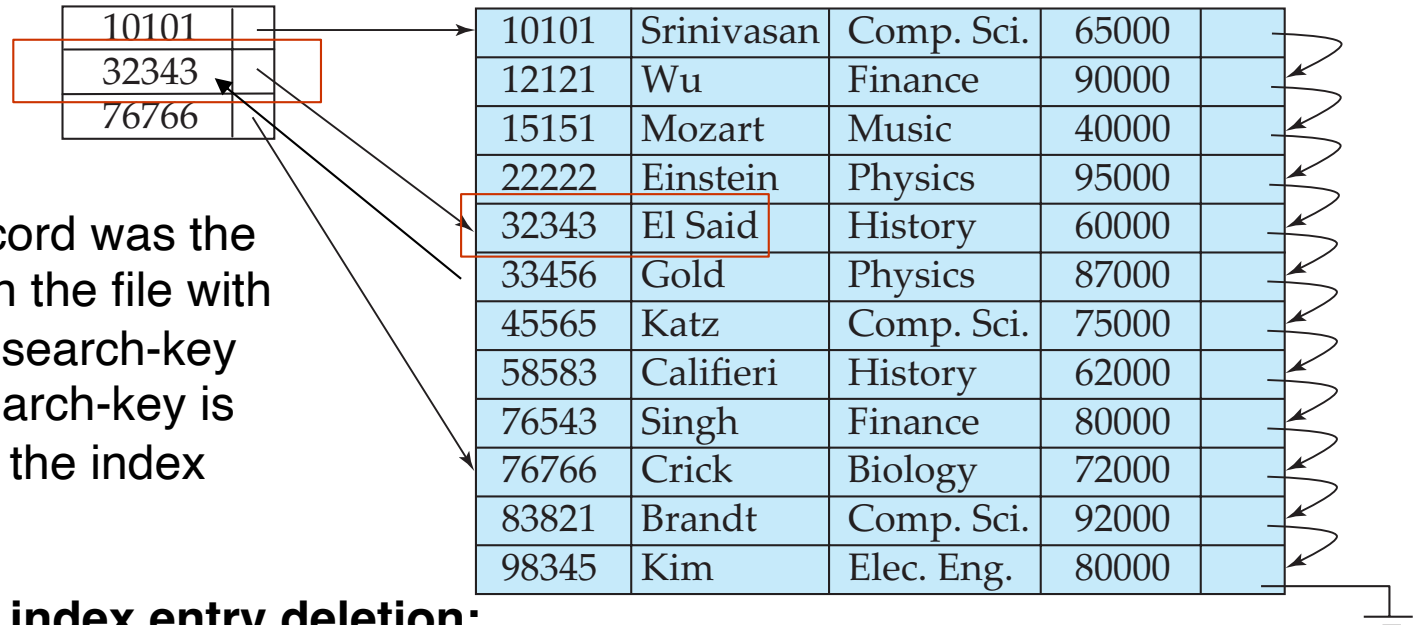


Multilevel Index (Cont.)





Index Update: Deletion



- If deleted record was the only record in the file with its particular search-key value, the search-key is deleted from the index also.
- **Single-level index entry deletion:**
 - **Dense indices** – deletion of search-key is similar to file record deletion.
 - **Sparse indices** –
 - if an entry for the search key exists in the index, it is deleted by replacing the entry in the index with the next search-key value in the file (in search-key order).
 - If the next search-key value already has an index entry, the entry is deleted instead of being replaced.



Index Update: Insertion

- **Single-level index insertion:**

- Perform a lookup using the search-key value of the record to be inserted.
- **Dense indices** – if the search-key value does not appear in the index, insert it
 - Indices are maintained as sequential files
 - Need to create space for new entry, overflow blocks may be required
- **Sparse indices** – if index stores an entry for each block of the file, no change needs to be made to the index unless a new block is created.
 - If a new block is created, the first search-key value appearing in the new block is inserted into the index.



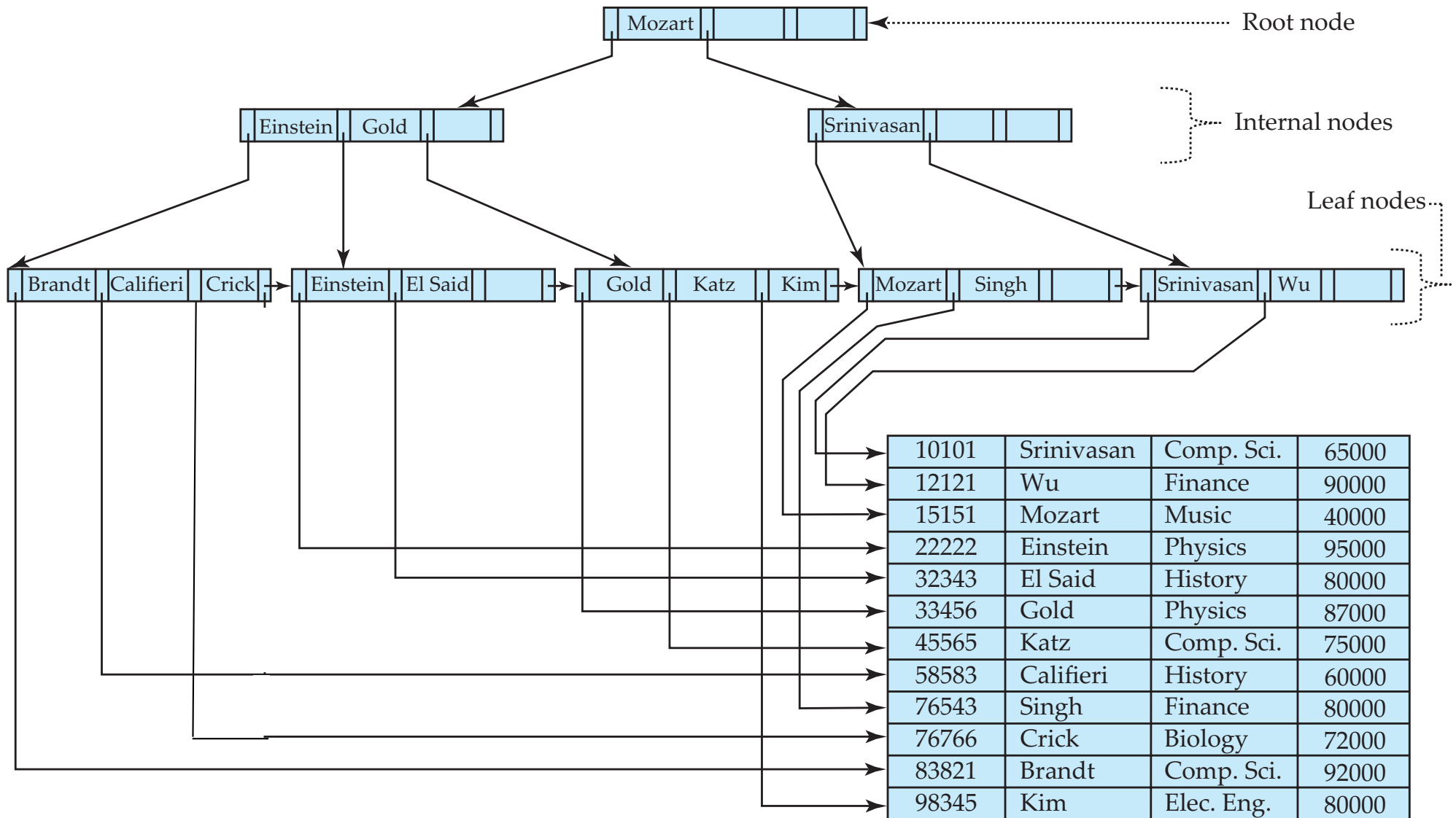
Indices on Multiple Keys

- **Composite search key**

- E.g., index on *instructor* relation on attributes (*name*, *ID*)
- Values are sorted lexicographically
 - E.g. (John, 12121) < (John, 13514) and (John, 13514) < (Peter, 11223)
- Can query on just *name*, or on (*name*, *ID*)



B⁺-Tree Index



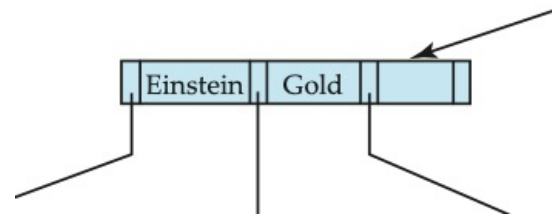


B⁺-Tree Index Files (Cont.)

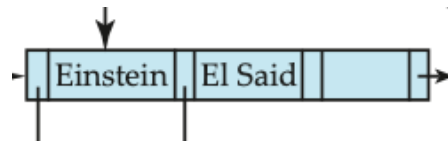
A B⁺-tree is a rooted tree satisfying the following properties:

- All paths from root to leaf are of the same length
- Each node that is not a root or a leaf has between $\lceil n/2 \rceil$ and n children.

- e.g., $n=4$



- A leaf node has between $\lceil (n-1)/2 \rceil$ and $n-1$ values



- Special cases:
 - If the root is not a leaf, it has at least 2 children.
 - If the root is a leaf (that is, there are no other nodes in the tree), it can have between 0 and $(n-1)$ values.



B⁺-Tree Node Structure

- Typical node

P_1	K_1	P_2	...	P_{n-1}	K_{n-1}	P_n
-------	-------	-------	-----	-----------	-----------	-------

- K_i are the search-key values
 - P_i are pointers to children (for non-leaf nodes) or pointers to records or buckets of records (for leaf nodes).
- The search-keys in a node are ordered

$$K_1 < K_2 < K_3 < \dots < K_{n-1}$$

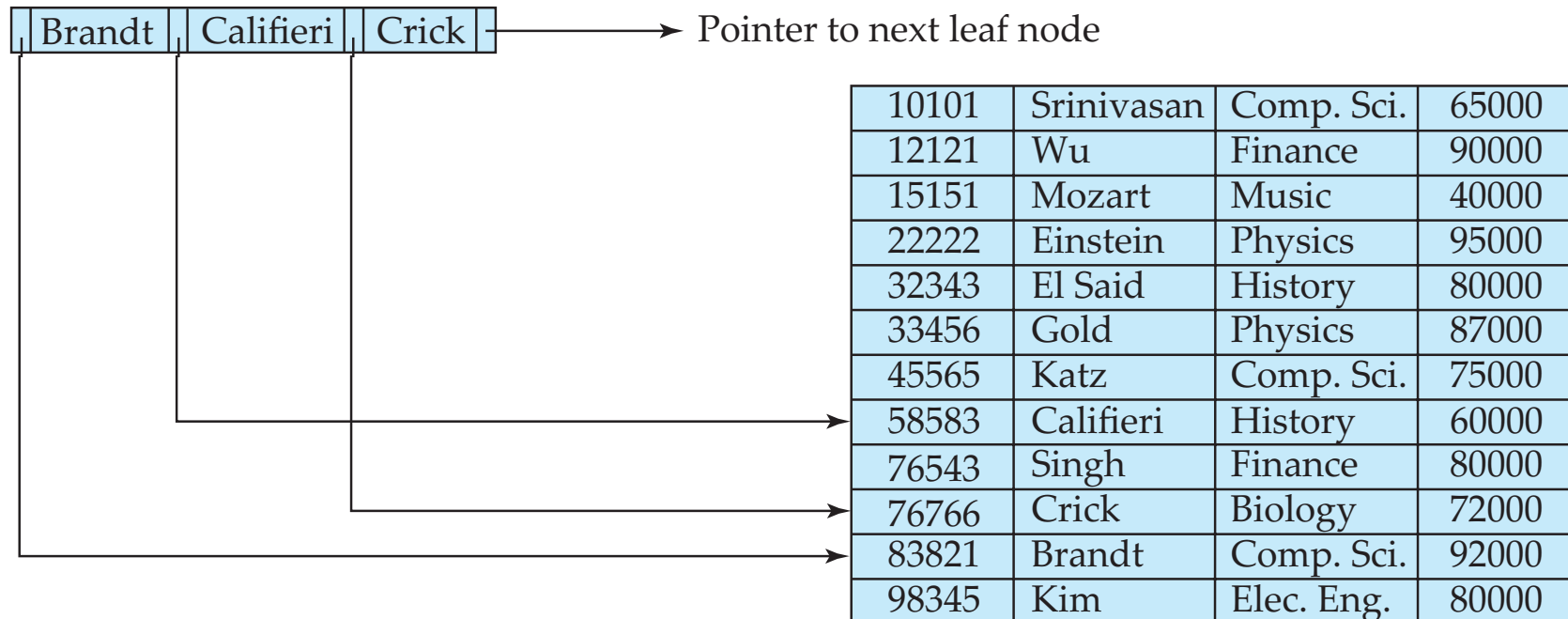
(Initially assume no duplicate keys, address duplicates later)



Leaf Nodes in B⁺-Trees

Properties of a leaf node:

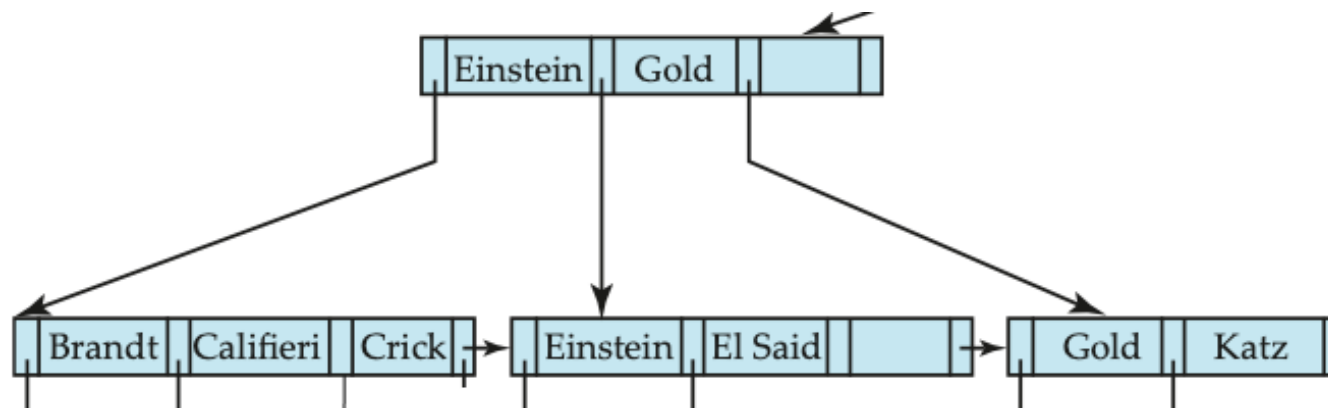
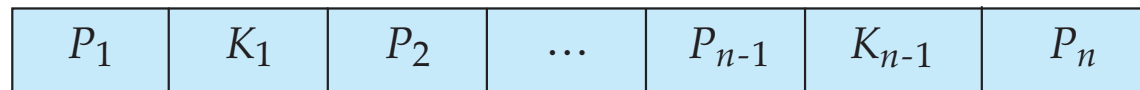
- For $i = 1, 2, \dots, n-1$, pointer P_i points to a file record with search-key value K_i
- If L_i, L_j are leaf nodes and $i < j$, L_i 's search-key values are less than or equal to L_j 's search-key values
- P_n points to next leaf node in search-key order
leaf node





Non-Leaf Nodes in B+-Trees

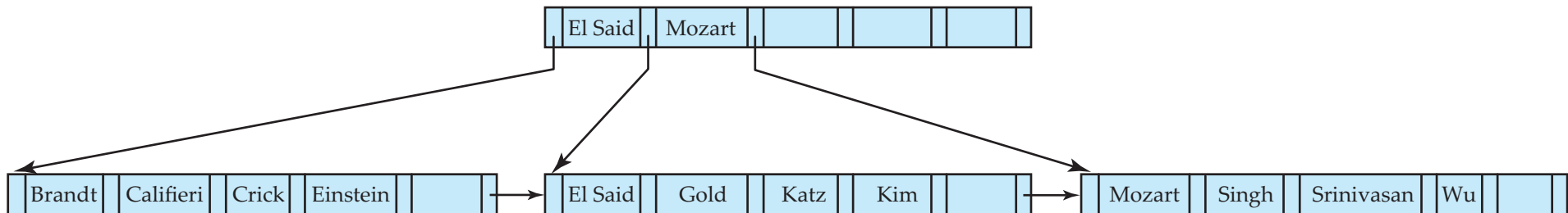
- Non leaf nodes form a multi-level sparse index on the leaf nodes. For a non-leaf node with m pointers:
 - All the search-keys in the subtree to which P_1 points are less than K_1
 - For $2 \leq i \leq n - 1$, all the search-keys in the subtree to which P_i points have values greater than or equal to K_{i-1} and less than K_i
 - All the search-keys in the subtree to which P_n points have values greater than or equal to K_{n-1}
 - General structure





Example of B⁺-tree

- B⁺-tree for *instructor* file ($n = 6$)



- Leaf nodes must have between 3 and 5 values ($\lceil (n-1)/2 \rceil$ and $n-1$, with $n = 6$).
- Non-leaf nodes other than root must have between 3 and 6 children ($\lceil n/2 \rceil$ and n with $n = 6$).
- Root must have at least 2 children.



Observations about B⁺-trees

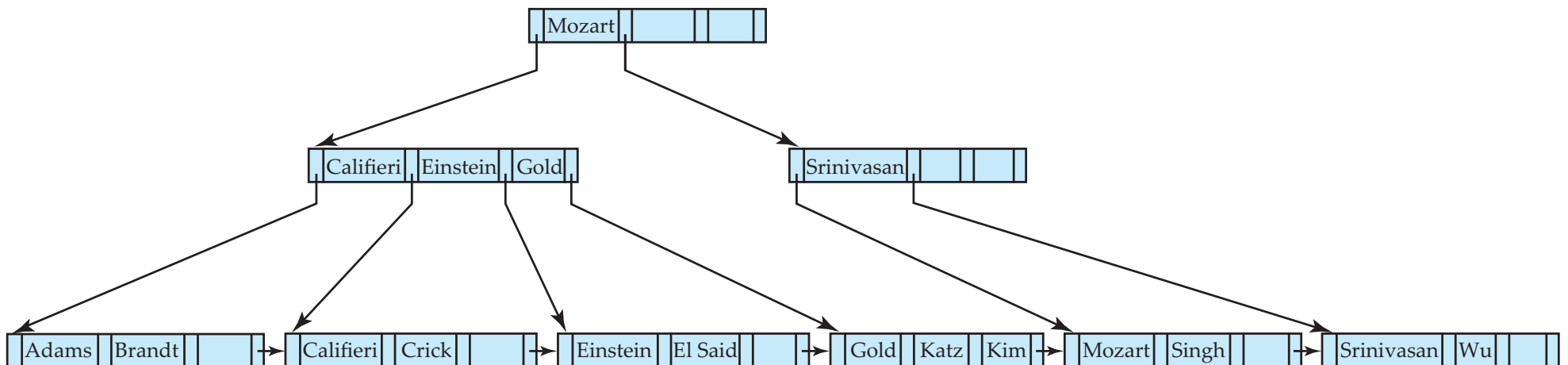
- Since the inter-node connections are done by pointers, “logically” close blocks need not be “physically” close.
- The non-leaf levels of the B⁺-tree form a hierarchy of sparse indices.
- The B⁺-tree contains a relatively small number of levels
 - Level below root has at least $2 * \lceil n/2 \rceil$ values
 - Next level has at least $2 * \lceil n/2 \rceil * \lceil n/2 \rceil$ values
 - .. etc.
- If there are K search-key values in the file, the tree height is no more than $\lceil \log_{\lceil n/2 \rceil}(K) \rceil$
- thus searches can be conducted efficiently.
- Insertions and deletions to the main file can be handled efficiently, as the index can be restructured in logarithmic time (as we shall see).



Queries on B⁺-Trees

function *find*(*V*)

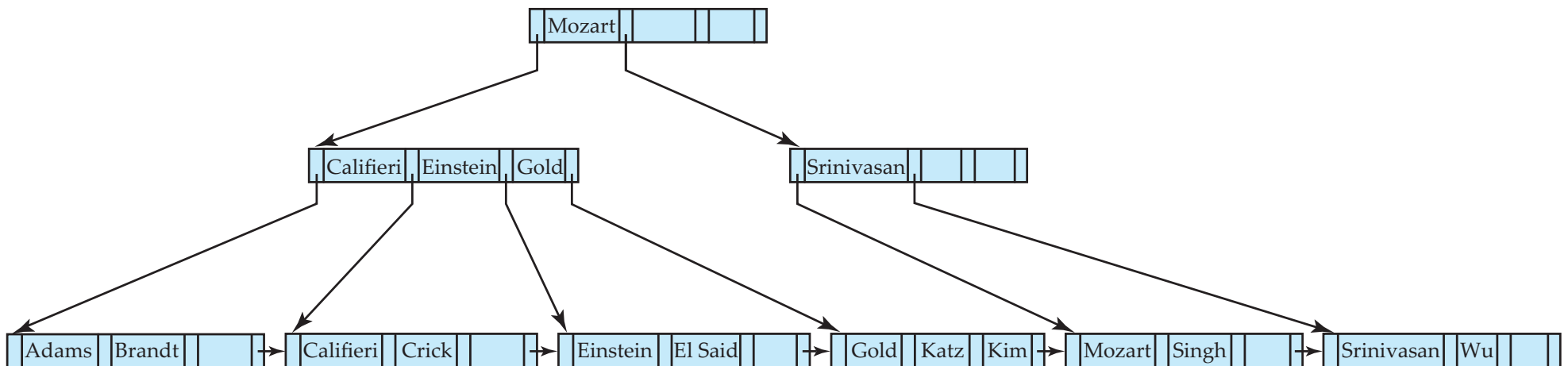
1. *C* = *root*
2. **while** (*C* is not a leaf node)
 1. Let *i* be least number s.t. $V \leq K_i$.
 2. **if** there is no such number *i* **then**
 3. Set *C* = *last non-null pointer in C*
 4. **else if** ($V = C.K_i$) Set *C* = *P_{i+1}*
 5. **else set** *C* = *C.P_i*
3. **if** for some *i*, $K_i = V$ **then** return *C.P_i*
4. **else** return null /* no record with search-key value *v* exists. */





Queries on B⁺-Trees (Cont.)

- **Range queries** find all records with search key values in a given range (lb, ub)
 - it first traverses to a leaf in a manner similar to $find(lb)$
 - It then steps through records in that and subsequent leaf nodes collecting pointers to all records with key values $C.K_i$ s.t. $lb \leq C.K_i \leq ub$ into a set $resultSet$.
 - The function stops when $C.K_i > ub$, or there are no more keys in the tree.





Queries on B⁺-Trees (Cont.)

- If there are K search-key values in the file, the height of the tree is no more than $\lceil \log_{\lceil n/2 \rceil}(K) \rceil$.
- A node is generally the same size as a disk block, typically 4 kilobytes
 - and n is typically around 100 (40 bytes per index entry).
- With 1 million search key values and $n = 100$
 - at most $\log_{50}(1,000,000) = 4$ nodes are accessed in a lookup traversal from root to leaf.
 - every node access may need a disk I/O, costing around 20 milliseconds



Updates on B⁺-Trees: Insertion

Assume record already added to the file. Let

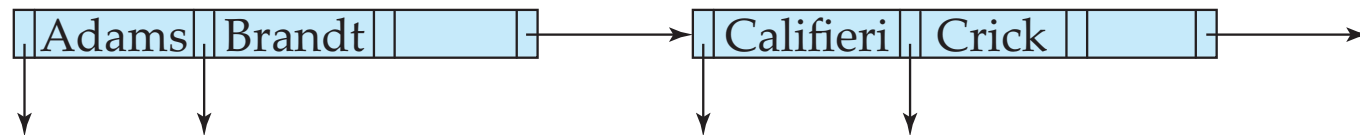
- pr be pointer to the record, and let
- v be the search key value of the record

1. Find the leaf node in which the search-key value would appear
 1. If there is room in the leaf node, insert (v, pr) pair in the leaf node
 2. Otherwise, split the node (along with the new (v, pr) entry) as discussed in the next slide, and propagate updates to parent nodes.



Updates on B⁺-Trees: Insertion (Cont.)

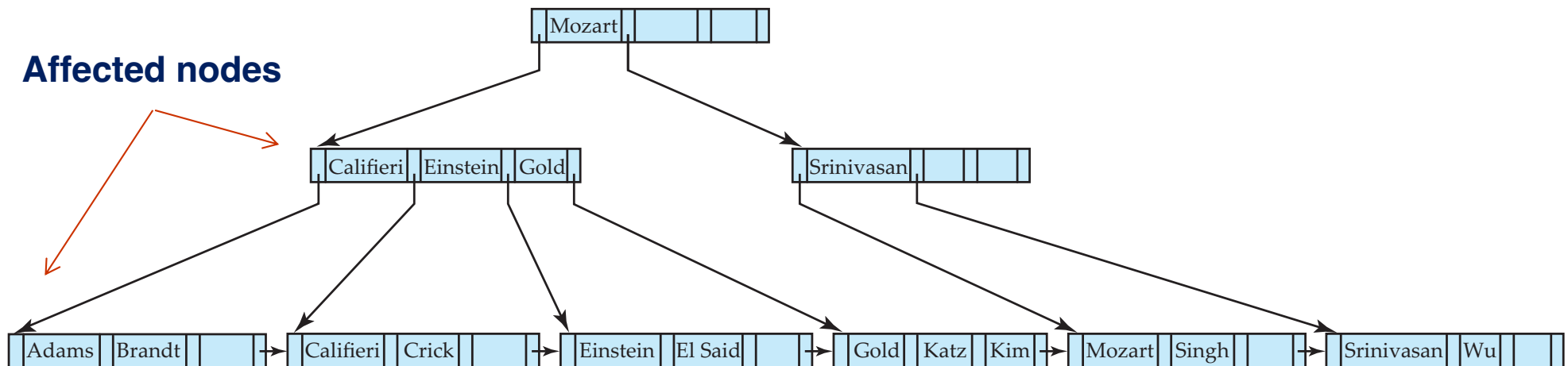
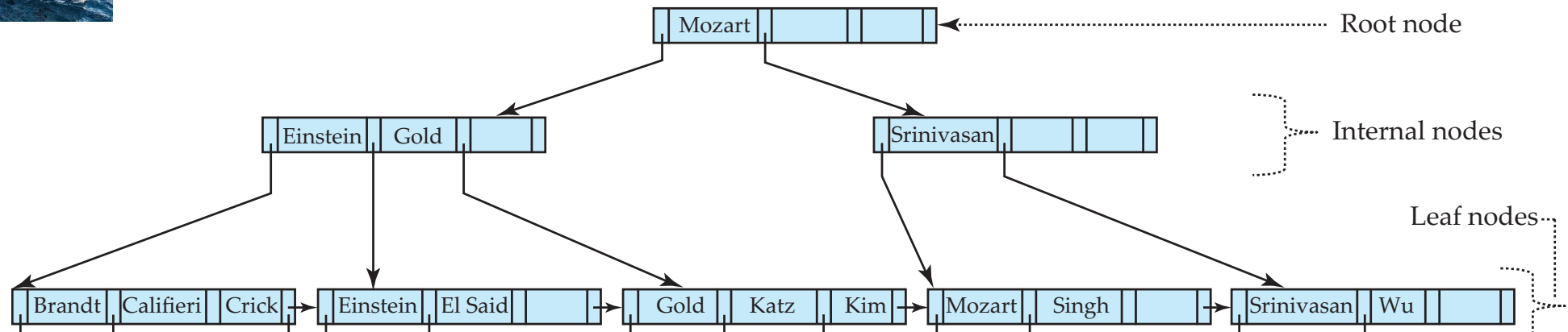
- Splitting a leaf node:
 - take the n (search-key value, pointer) pairs (including the one being inserted) in sorted order. Place the first $\lceil n/2 \rceil$ in the original node, and the rest in a new node.
 - let the new node be p , and let k be the least key value in p . Insert (k,p) in the parent of the node being split.
 - If the parent is full, split it and **propagate** the split further up.
- Splitting of nodes proceeds upwards till a node that is not full is found.
 - In the worst case the root node may be split increasing the height of the tree by 1.



Result of splitting node containing Brandt, Califieri and Crick on inserting Adams
Next step: insert entry with (Califieri, pointer-to-new-node) into parent



B⁺-Tree Insertion

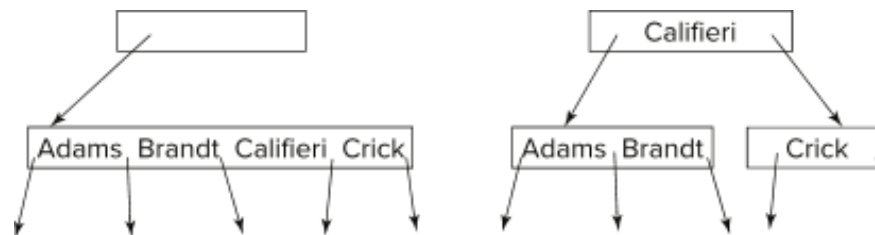


B⁺-Tree before and after insertion of “Adams”



Insertion in B⁺-Trees (Cont.)

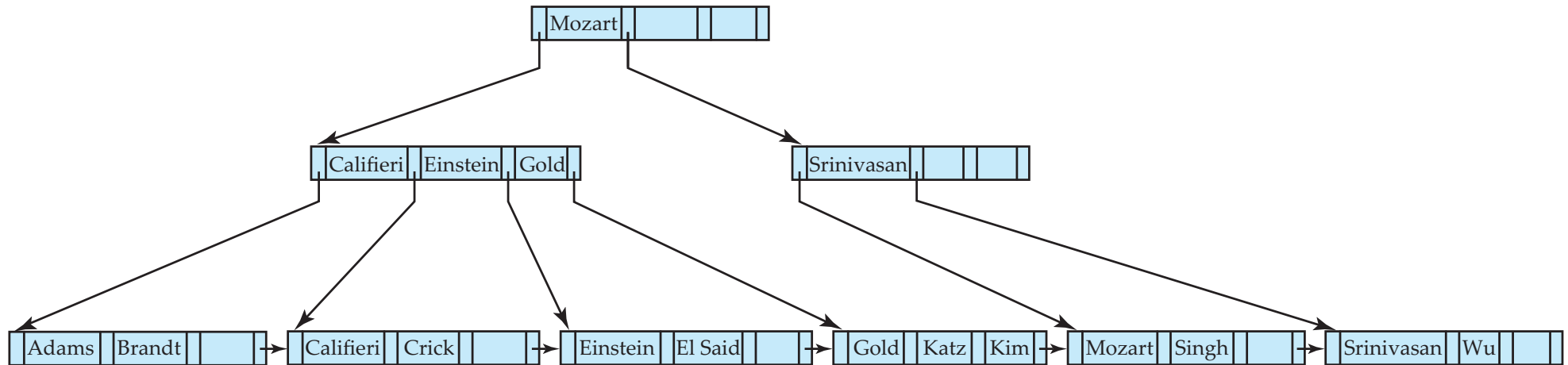
- Splitting a non-leaf node: when inserting (k,p) into an already full internal node N
 - Copy N to an in-memory area M with space for n+1 pointers and n keys
 - Insert (k,p) into M
 - Copy $P_1, K_1, \dots, K_{\lceil n/2 \rceil - 1}, P_{\lceil n/2 \rceil}$ from M back into node N
 - Copy $P_{\lceil n/2 \rceil + 1}, K_{\lceil n/2 \rceil + 1}, \dots, K_n, P_{n+1}$ from M into newly allocated node N'
 - Insert $(K_{\lceil n/2 \rceil}, N')$ into parent N
- Example



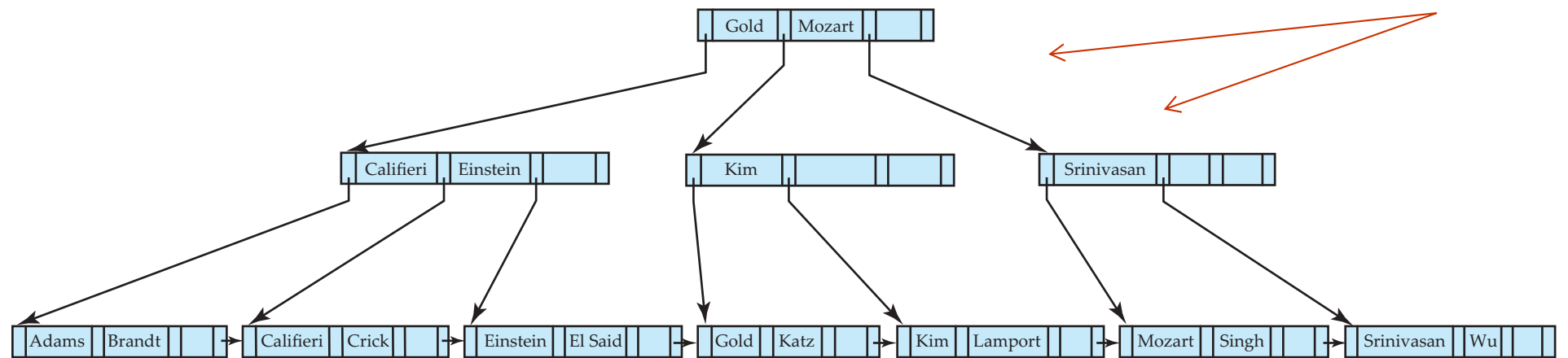
- **Read pseudocode in book!**



B⁺-Tree Insertion



B⁺-Tree before and after insertion of “Lampport”

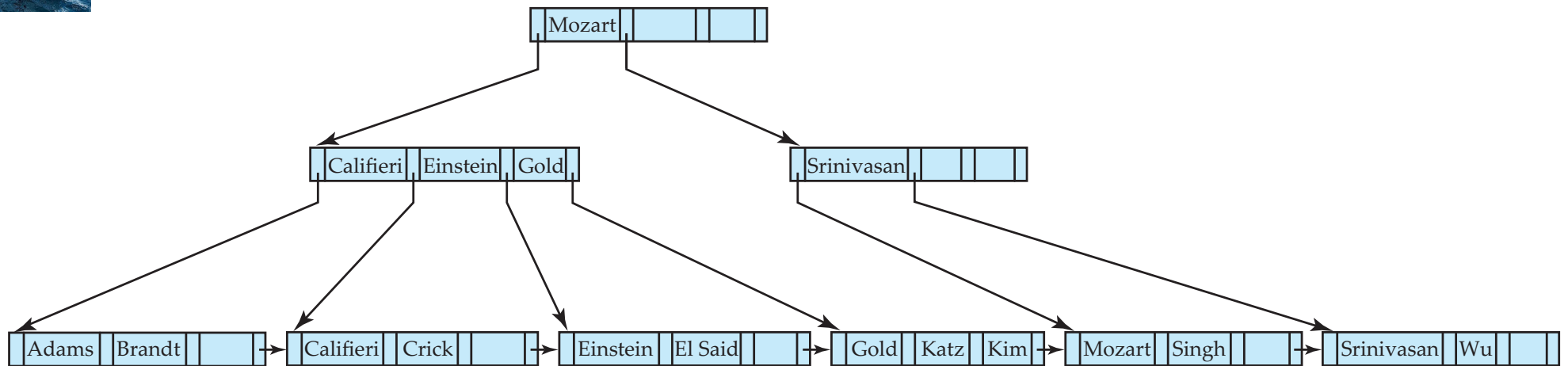


Affected nodes

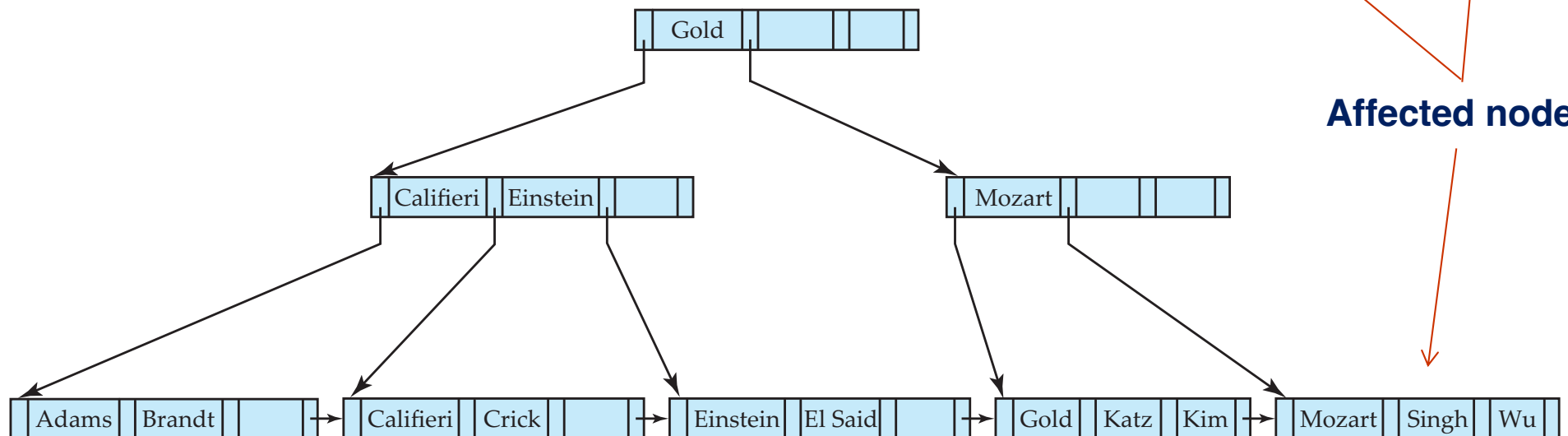
Affected nodes



Examples of B⁺-Tree Deletion



Before and after deleting “Srinivasan”

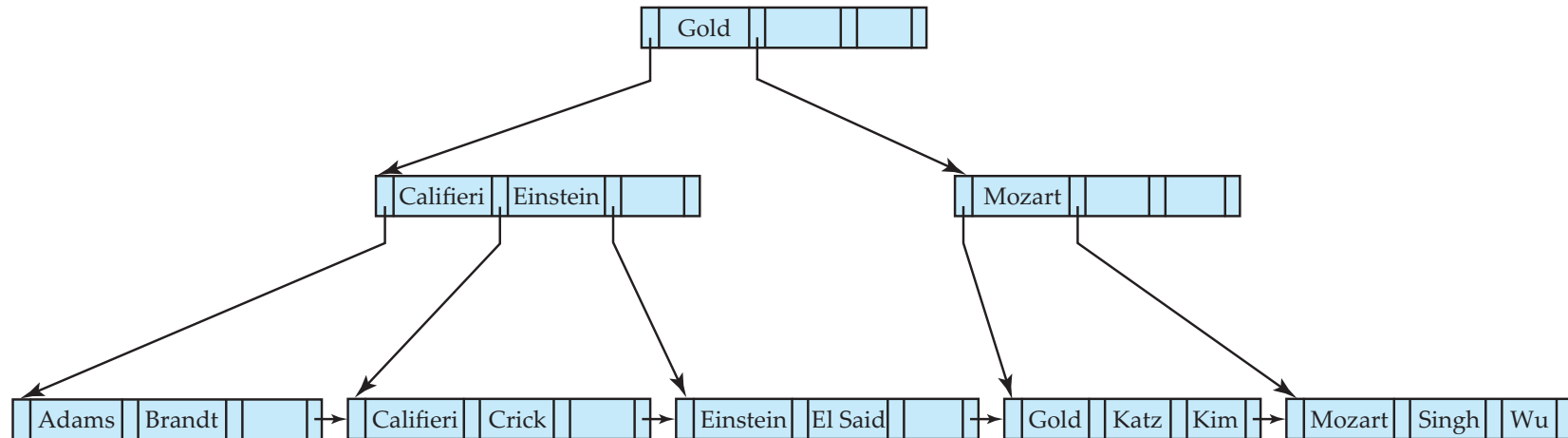


Affected nodes

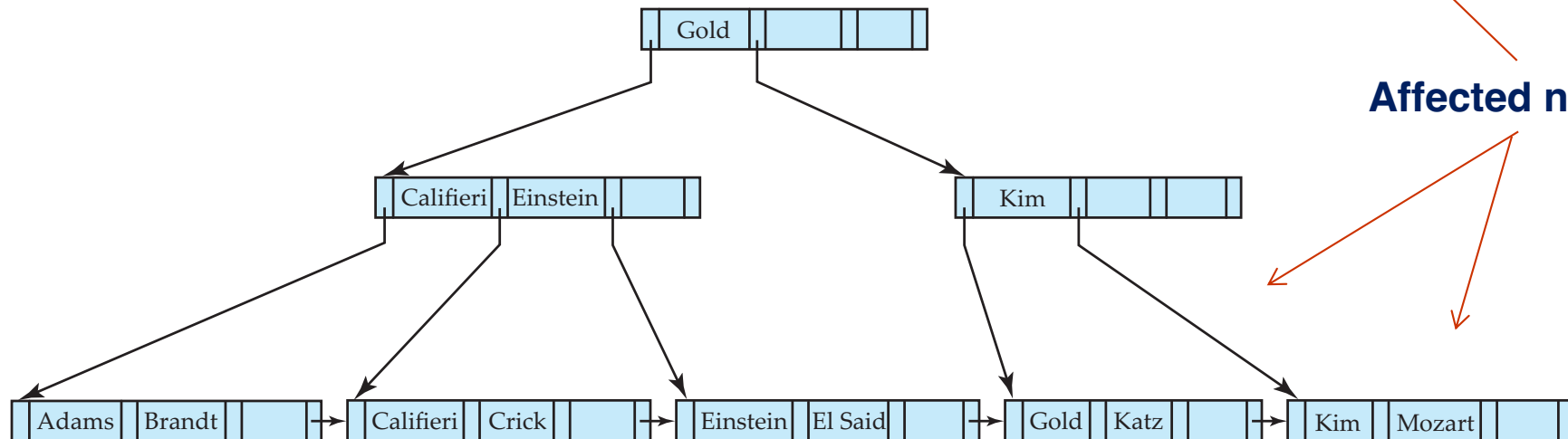
- Deleting “Srinivasan” causes **merging** of under-full leaves



Examples of B⁺-Tree Deletion (Cont.)



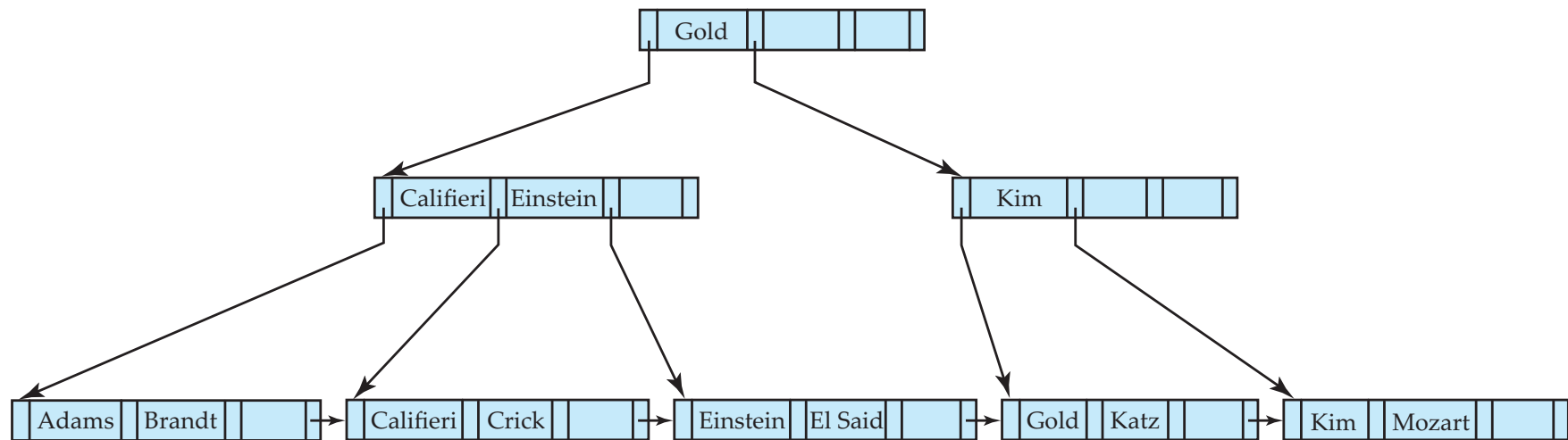
Before and after deleting “Singh” and “Wu”



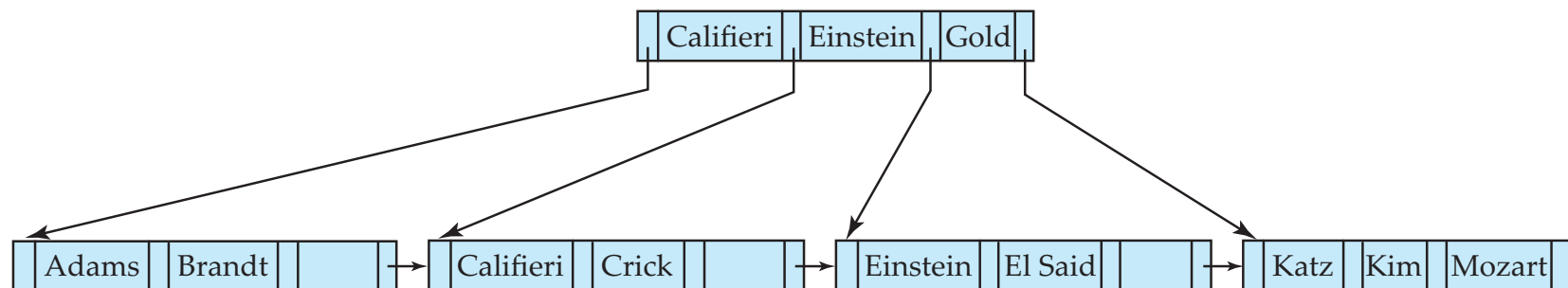
- Leaf containing Singh and Wu became underfull, and **borrowed a value** Kim from its left sibling
- Search-key value in the parent changes as a result



Example of B⁺-tree Deletion (Cont.)



Before and after deletion of “Gold”



- Node with Gold and Katz became underfull, and was merged with its sibling
- Parent node becomes underfull, and is merged with its sibling
 - Value separating two nodes (at the parent) is pulled down when merging
- Root node then has only one child, and is deleted



Updates on B⁺-Trees: Deletion

Assume record already deleted from file. Let V be the search key value of the record, and Pr be the pointer to the record.

- Remove (Pr, V) from the leaf node
- If the node has too few entries due to the removal, and the entries in the node and a sibling fit into a single node, then **merge siblings**:
 - Insert all the search-key values in the two nodes into a single node (the one on the left), and delete the other node.
 - Delete the pair (K_{i-1}, P_i) , where P_i is the pointer to the deleted node, from its parent, recursively using the above procedure.



Updates on B⁺-Trees: Deletion

- Otherwise, if the node has too few entries due to the removal, but the entries in the node and a sibling do not fit into a single node, then **redistribute pointers**:
 - Redistribute the pointers between the node and a sibling such that both have more than the minimum number of entries.
 - Update the corresponding search-key value in the parent of the node.
- The node deletions may cascade upwards till a node which has $\lceil n/2 \rceil$ or more pointers is found.
- If the root node has only one pointer after deletion, it is deleted and the sole child becomes the root.



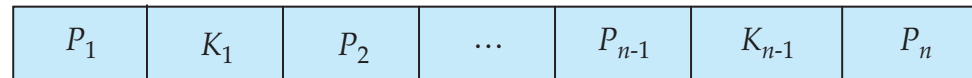
Complexity of Updates

- Cost (in terms of number of I/O operations) of insertion and deletion of a single entry proportional to height of the tree
 - With K entries and maximum fanout of n , worst case complexity of insert/delete of an entry is $O(\log_{\lceil n/2 \rceil}(K))$
- In practice, number of I/O operations is less:
 - Internal nodes tend to be in buffer
 - Splits/merges are rare, most insert/delete operations only affect a leaf node

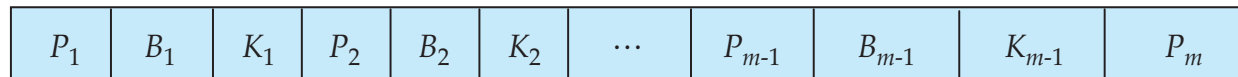


B-Tree Index Files

- Similar to B+-tree, but B-tree allows search-key values to appear only once; eliminates redundant storage of search keys.
- Search keys in nonleaf nodes appear nowhere else in the B-tree; an additional pointer field for each search key in a nonleaf node must be included.
- Generalized B-tree leaf node



(a)

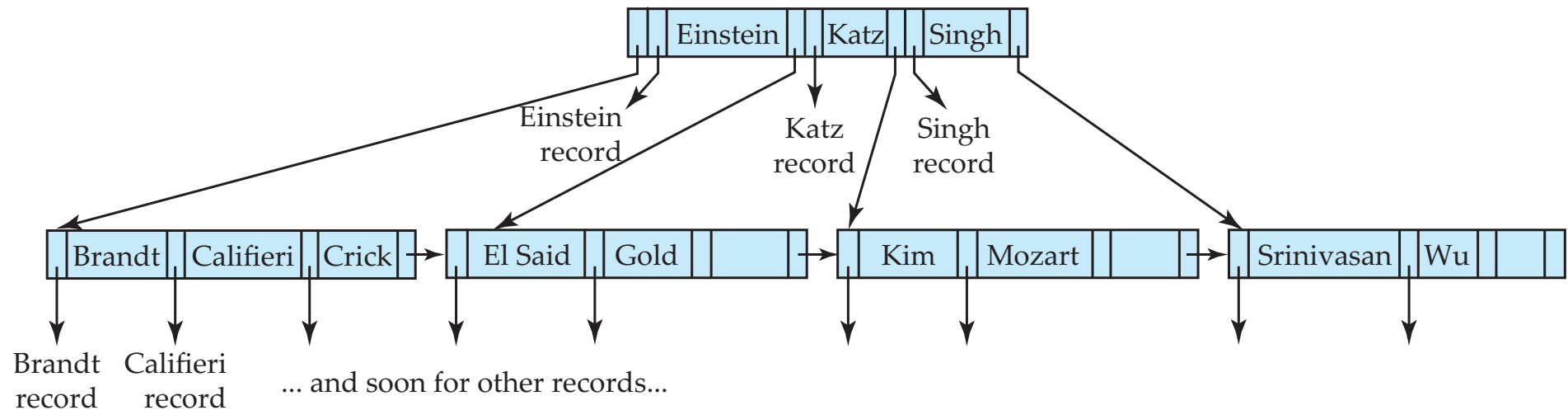


(b)

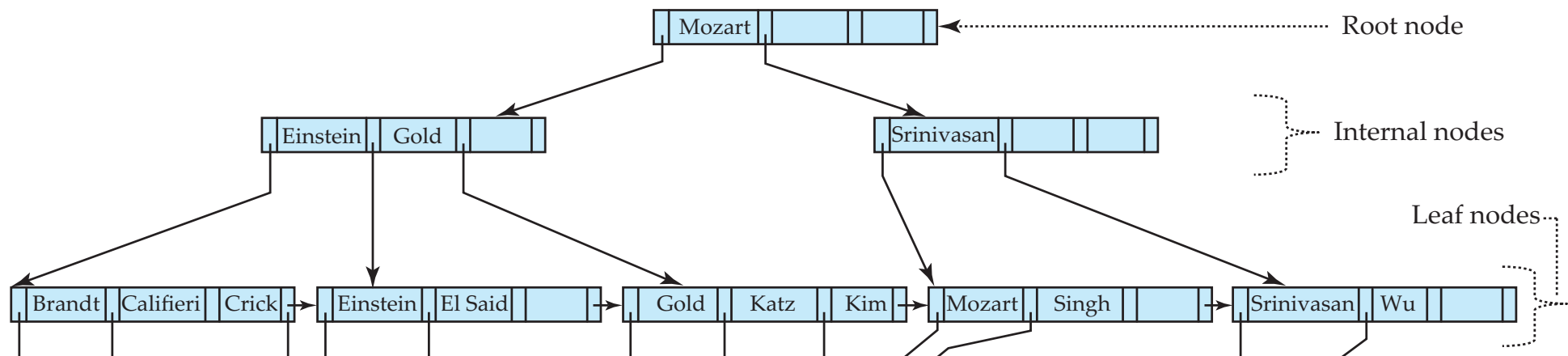
- Nonleaf node – pointers B_i are the bucket or file record pointers.



B-Tree Index File Example



B-tree (above) and B+-tree (below) on same data





Hashing



Static Hashing

- A **bucket** is a unit of storage containing one or more entries (a bucket is typically a disk block).
 - we obtain the bucket of an entry from its search-key value using a **hash function**
- Hash function h is a function from the set of all search-key values K to the set of all bucket addresses B .
- Hash function is used to locate entries for access, insertion as well as deletion.
- Entries with different search-key values may be mapped to the same bucket; thus entire bucket has to be searched sequentially to locate an entry.
- In a **hash index**, buckets store entries with pointers to records
- In a **hash file-organization** buckets store records



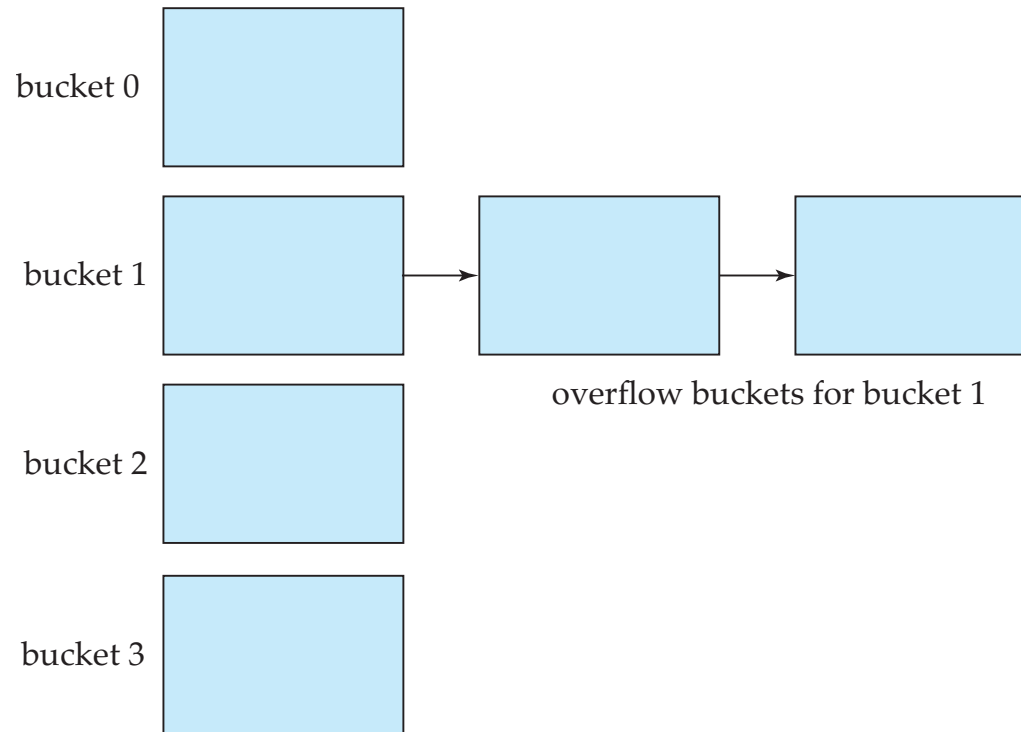
Handling of Bucket Overflows

- Bucket overflow can occur because of
 - Insufficient buckets
 - Skewness in distribution of records. This can occur due to two reasons:
 - multiple records have same search-key value
 - chosen hash function produces non-uniform distribution of key values
- Although the probability of bucket overflow can be reduced, it cannot be eliminated; it is handled by using ***overflow buckets***.



Handling of Bucket Overflows (Cont.)

- **Overflow chaining** – the overflow buckets of a given bucket are chained together in a linked list.
- Above scheme is called **closed addressing** (also called **closed hashing** or **open hashing** depending on the book you use)





Example of Hash File Organization

Hash file organization of *instructor* file, using *dept_name* as key.

E.g.

$h(\text{Music}) = 1$

$h(\text{History}) = 2$

$h(\text{Physics}) = 3$

$h(\text{Elec. Eng.}) = 3$

bucket 0

bucket 1

15151	Mozart	Music	40000

bucket 2

32343	El Said	History	80000
58583	Califieri	History	60000

bucket 3

22222	Einstein	Physics	95000
33456	Gold	Physics	87000
98345	Kim	Elec. Eng.	80000

bucket 4

12121	Wu	Finance	90000
76543	Singh	Finance	80000

bucket 5

76766	Crick	Biology	72000

bucket 6

10101	Srinivasan	Comp. Sci.	65000
45565	Katz	Comp. Sci.	75000
83821	Brandt	Comp. Sci.	92000

bucket 7



Example of Hash Index

bucket 0

76766	

bucket 1

45565	
76543	

bucket 2

22222	

bucket 3

10101	

bucket 4

bucket 5

15151	
33456	

58583	
98345	

bucket 6

83821	

bucket 7

12121	
32343	

76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
45565	Katz	Comp. Sci.	75000
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76543	Singh	Finance	80000
32343	El Said	History	60000
58583	Califieri	History	62000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
33465	Gold	Physics	87000

hash index on *instructor*, on attribute *ID*



Deficiencies of Static Hashing

- In static hashing, function h maps search-key values to a fixed set of B of bucket addresses. Databases grow or shrink with time.
 - If initial number of buckets is too small, and file grows, performance will degrade due to too much overflows.
 - If space is allocated for anticipated growth, a significant amount of space will be wasted initially (and buckets will be underfull).
 - If database shrinks, again space will be wasted.
- One solution: periodic re-organization of the file with a new hash function
 - Expensive, disrupts normal operations
- Better solution: allow the number of buckets to be modified dynamically.



Dynamic Hashing

- Periodic rehashing
 - If number of entries in a hash table becomes (say) 1.5 times size of hash table,
 - create new hash table of size (say) 2 times the size of the previous hash table
 - Rehash all entries to new table
- Linear Hashing
 - Do rehashing in an incremental manner
- Extendable Hashing
 - Tailored to disk based hashing, with buckets shared by multiple hash values
 - Doubling of # of entries in hash table, without doubling # of buckets



Comparison of Ordered Indexing and Hashing

- Cost of periodic re-organization
- Relative frequency of insertions and deletions
- Is it desirable to optimize average access time at the expense of worst-case access time?
- Expected type of queries:
 - Hashing is generally better at retrieving records having a specified value of the key.
 - If range queries are common, ordered indices are to be preferred
- In practice:
 - PostgreSQL supports hash indices, but discourages use due to poor performance
 - Oracle supports static hash organization, but not hash indices
 - SQLServer supports only B⁺-trees



Multiple-Key Access

- Use multiple indices for certain types of queries.
- Example:
select *ID*
from *instructor*
where *dept_name* = “Finance” **and** *salary* = 80000
- Possible strategies for processing query using indices on single attributes:
 1. Use index on *dept_name* to find instructors with department name Finance; test *salary* = 80000
 2. Use index on *salary* to find instructors with a salary of \$80000; test *dept_name* = “Finance”.
 3. Use *dept_name* index to find pointers to all records pertaining to the “Finance” department. Similarly use index on *salary*. Take intersection of both sets of pointers obtained.



Indices on Multiple Keys

- **Composite search keys** are search keys containing more than one attribute
 - E.g., (*dept_name*, *salary*)
- Lexicographic ordering: $(a_1, a_2) < (b_1, b_2)$ if either
 - $a_1 < b_1$, or
 - $a_1 = b_1$ and $a_2 < b_2$



Indices on Multiple Attributes

Suppose we have an index on combined search-key
(*dept_name*, *salary*).

- With the **where** clause
 where *dept_name* = “Finance” **and** *salary* = 80000
the index on (*dept_name*, *salary*) can be used to fetch only records that satisfy both conditions.
 - Using separate indices is less efficient — we may fetch many records (or pointers) that satisfy only one of the conditions.
- Can also efficiently handle
 where *dept_name* = “Finance” **and** *salary* < 80000
- But cannot efficiently handle
 where *dept_name* < “Finance” **and** *balance* = 80000
 - May fetch many records that satisfy the first but not the second condition



Creation of Indices

- Example
 - create index** *takes_pk* **on** *takes (ID, course_ID, year, semester, section)*
 - drop index** *takes_pk*
- Most database systems allow specification of type of index, and clustering.
- Indices on primary key created automatically by all databases
- Indices can greatly speed up lookups, but impose cost on updates
 - Index tuning assistants/wizards supported on several databases to help choose indices, based on query and update workload



Index Definition in SQL

- Create an index

create index <index-name> **on** <relation-name>
(<attribute-list>)

E.g.,: **create index** *b-index* **on** *branch(branch_name)*

- To drop an index

drop index <index-name>

- Most database systems allow specification of type of index, and clustering.



End of Chapter 14